

PHYS505 Recitation — Feb 23, 2026

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Problem 4.7

Localized charge distribution representative of the $m = \pm 1$ states of the 2p level in hydrogen

$$\rho(\vec{r}) = \frac{1}{64\pi} r^2 e^{-r} \sin^2 \theta$$

- (a) Find the multipole expansion of the corresponding potential including all non-0 moments, and a finite Legendre expansion at large r .
(b) Find the potential everywhere and show that near the origin, to order r^2 ,

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{1}{4} - \frac{r^2}{120} P_2(\cos \theta) \right]$$

- (c) Find the interaction energy for a nucleus at the origin with quadrupole moment $Q = 10^{-28} \text{ m}^2$, with charge e and Bohr radius

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} = 5.29 \times 10^{-11} \text{ m}$$

and express the answer as a frequency $f = W/h$.

Note that the wavefunction ψ for the $n = 2$, $l = 1$, $m = \pm 1$ states in the hydrogen atom is

$$\psi(r, \theta, \phi) = \mp \frac{1}{8\sqrt{\pi}a_0^3} \frac{r}{a_0} e^{-r/(2a_0)} \sin \theta e^{\pm i\phi}$$

The corresponding probability density $|\psi|^2$ results in the expression we are given with $a_0 = 1$ (r is measured in units of a_0).

- (a) We want to find the multipole expansion

$$\Phi(\vec{r}) = \frac{1}{\epsilon_0} \sum_{l,m} \frac{q_{lm}}{2l+1} \frac{Y_{lm}(\theta, \phi)}{r^{l+1}}$$

with the multipole moments

$$q_{lm} = \int d^3x Y_{lm}^*(\theta, \phi) r^l \rho(\vec{x}) = \int \int r^2 d\Omega dr Y_{lm}^*(\theta, \phi) r^l \rho(\vec{x})$$

We can use the orthogonality of the spherical harmonics over the unit sphere (using Dirac notation, not typical in this context)

$$\int d\Omega Y_{l'm'}^*(\theta, \phi) Y_{lm}(\theta, \phi) = \langle Y_{l'm'} | Y_{lm} \rangle = \delta_{ll'} \delta_{mm'}$$

if we express the angular part of ρ in terms of them. The charge distribution has azimuthal symmetry, so its angular expansion includes only the spherical harmonics with $m = 0$

$$\rho(r, \theta) = f(r) \sum_l c_l |Y_{l0}\rangle$$

$$f(r) = \frac{1}{64\pi} r^2 e^{-r}$$

The $m = 0$ terms are proportional to the Legendre polynomials $P_l(\cos \theta)$ and we can find the coefficients by inspection.

$$\sin^2 \theta = 1 - \cos^2 \theta$$

is a 2nd order polynomial in $\cos \theta$, so we look at terms up to P_2

$$P_0 = 1$$

$$P_1(\cos \theta) = \cos \theta$$

$$P_2(\cos \theta) = \frac{3}{2} \cos^2 \theta - \frac{1}{2}$$

$$1 - \cos^2 \theta = \frac{2}{3} P_0(\cos \theta) - \frac{2}{3} P_2(\cos \theta)$$

and the corresponding spherical harmonics are

$$Y_{00} = \frac{1}{\sqrt{4\pi}} P_0$$

$$Y_{20} = \sqrt{\frac{5}{4\pi}} P_2$$

$$1 - \cos^2 \theta = \frac{2}{3} \sqrt{4\pi} \left(Y_{00} - \frac{1}{\sqrt{5}} Y_{20} \right)$$

so we have the expression we need with the coefficients

$$c_0 = \frac{2}{3} \sqrt{4\pi}$$

$$c_2 = -\frac{2}{3} \sqrt{\frac{4\pi}{5}}$$

and can compute the multipole moments

$$\begin{aligned}
q_{lm} &= \int [r^2 d\Omega dr] Y_{lm}^*(\theta, \phi) r^l \left[f(r) \sum_{l'} c_{l'} Y_{l'0}(\theta, \phi) \right] \\
&= \left[\int dr r^{l+2} f(r) \right] \sum_{l'} c_{l'} \langle Y_{lm} | Y_{l'0} \rangle \\
&= \left[\frac{1}{64\pi} \int dr r^{l+4} e^{-r} \right] c_l \delta_{m0}
\end{aligned}$$

The radial integral is straightforward

$$\begin{aligned}
I_0(\alpha) &= \int_0^\infty e^{-\alpha u} du = \frac{1}{\alpha} \\
\left(\frac{d}{d\alpha} \right)^n I_0(\alpha) &= (-1)^n \int_0^\infty u^n e^{-\alpha u} du = (-1)^n n! \alpha^{-(n+1)} \\
\int_0^\infty r^n e^{-\alpha r} dr &= \frac{n!}{\alpha^{n+1}}
\end{aligned}$$

so we have

$$\begin{aligned}
q_{l0} &= \frac{(l+4)!}{64\pi} c_l, \quad l > 0 \\
q_{00} &= \frac{4!}{64\pi} c_0 = \frac{1}{2\sqrt{\pi}} \\
q_{20} &= \frac{6!}{64\pi} c_2 = -3\sqrt{\frac{5}{\pi}}
\end{aligned}$$

the multipole expansion is

$$\begin{aligned}
\Phi &= \frac{1}{\epsilon_0} \left(q_{00} \frac{Y_{00}}{r} + \frac{q_{20}}{5} \frac{Y_{20}}{r^3} \right) \\
\Phi(r, \theta, \phi) &= \frac{1}{\epsilon_0} \frac{1}{\sqrt{4\pi}} \left[\frac{Y_{00}(\theta, \phi)}{r} - \frac{6}{\sqrt{5}} \frac{Y_{20}(\theta, \phi)}{r^3} \right]
\end{aligned}$$

and the Legendre expansion is

$$\Phi(r, \theta) = \frac{1}{4\pi\epsilon_0} \left[\frac{P_0(\cos \theta)}{r} - 6 \frac{P_2(\cos \theta)}{r^3} \right] = \frac{1}{4\pi\epsilon_0 r} \left[1 - \frac{3}{r^2} (3 \cos^2 \theta - 1) \right]$$

These expressions are valid *outside* the source region, that is, $r > R$ for some radius R beyond which ρ is negligible. While there is no sharp boundary in this problem, the charge density decays pretty fast due to the exponential factor. You can check that $r^2 e^{-r}$ peaks at $r = 2$ where it's about 0.54 and is down 3 orders of magnitude by $r = 10$.

(b) The expansion we found in part (a) is *not* valid *inside* the source region (near the origin, $r < 1$). To find a more general result, we can follow similar steps to the derivation of the multipole expansion but without assuming $r > r'$

$$\Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x'$$

We can expand the $1/r$ part in spherical harmonics but due to azimuthal symmetry, only the $m = 0$ terms are relevant

$$\begin{aligned} \frac{1}{|\vec{x} - \vec{x}'|} &= \frac{4\pi}{r_{>}} \sum_{l,m} \frac{1}{2l+1} \left(\frac{r_{<}}{r_{>}} \right)^l Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \\ &= \sum_l \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos \theta') P_l(\cos \theta) \end{aligned}$$

Using Dirac-like short-hand notation again, this time for the Legendre polynomials

$$\langle P_l | P_{l'} \rangle = \int_{-1}^1 dx P_l(x) P_{l'}(x) = \int_0^\pi d\theta \sin \theta P_l(\cos \theta) P_{l'}(\cos \theta) = \frac{2}{2l+1} \delta_{ll'}$$

and the Legendre expansion of the charge density from (a)

$$\rho(\vec{x}) = f(r) \left[\frac{2}{3} (|P_0\rangle - |P_2\rangle) \right]$$

we get

$$\begin{aligned} \Phi(r, \theta) &= \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} d\phi' \int dr' r'^2 f(r') \sum_l \frac{r_{<}^l}{r_{>}^{l+1}} \left[\frac{2}{3} (\langle P_0 | - \langle P_2 |) \right] |P_l\rangle P_l(\cos \theta) \\ &= \frac{1}{4\pi\epsilon_0} (2\pi) \frac{2}{3} \int_0^\infty dr' r'^2 f(r') \left[\frac{2}{r_{>}} P_0(\cos \theta) - \frac{2}{5} \frac{r_{<}^2}{r_{>}^3} P_2(\cos \theta) \right] \end{aligned}$$

where only two terms have survived the inner product of the Legendre polynomials. Note that we are integrating over the source x' , so r and θ are not part of the integral. In order to evaluate the radial integral correctly, we need to split the integration interval into two pieces, $r' < r$ (where $r_{<} = r'$, $r_{>} = r$) and $r' > r$ (with $r_{<} = r$, $r_{>} = r'$)

$$\Phi(r, \theta) = \frac{2}{3\epsilon_0} \left[\int_0^r dr' r'^2 f(r') \left(\frac{1}{r} P_0 - \frac{1}{5} \frac{r'^2}{r^3} P_2 \right) + \int_r^\infty dr' r'^2 f(r') \left(\frac{1}{r'} P_0 - \frac{1}{5} \frac{r^2}{r'^3} P_2 \right) \right]$$

The integrands are all of the kind $r^n e^{-r}$ but we need an integral formula that works for the interval $[b, \infty)$, where b is a positive constant. Using the same

method as before, we start by defining

$$I_b(\alpha) = \int_b^\infty e^{-\alpha u} du$$

$$\left(\frac{d}{d\alpha}\right)^n I_b(\alpha) = (-1)^n \int_b^\infty u^n e^{-\alpha u} du$$

and differentiating the formula for the result

$$I_b(\alpha) = \frac{e^{-\alpha b}}{\alpha}$$

$$\begin{aligned} \left(\frac{d}{d\alpha}\right)^n I_b(\alpha) &= \sum_{k=0}^n \frac{n!}{(n-k)!k!} \left[\left(\frac{d}{d\alpha}\right)^k \alpha^{-1}\right] \left[\left(\frac{d}{d\alpha}\right)^{n-k} e^{-\alpha b}\right] \\ &= \sum_{k=0}^n \frac{n!}{(n-k)!k!} [(-1)^k k! \alpha^{-1-k}] [(-\alpha)^{n-k} e^{-\alpha b}] \\ &= (-1)^n n! e^{-\alpha b} \sum_{k=0}^n \frac{\alpha^k}{k!} \alpha^{k-n-1} \\ \int_b^\infty r^n e^{-\alpha r} dr &= \frac{n! e^{-\alpha b}}{\alpha^{n+1}} \sum_{k=0}^n \frac{(\alpha b)^k}{k!} \end{aligned}$$

You can check that this has the correct limit as $b \rightarrow 0$ by inspection

$$\int_b^\infty r^n e^{-\alpha r} dr = \frac{n! e^{-\alpha b}}{\alpha^{n+1}} \left[1 + \alpha b + \frac{(\alpha b)^2}{2} \dots + \frac{(\alpha b)^n}{n!}\right]$$

The other type of integral we need can now be written in terms of the two we already have

$$\begin{aligned} \int_0^b r^n e^{-\alpha r} dr &= \int_0^\infty r^n e^{-\alpha r} dr - \int_b^\infty r^n e^{-\alpha r} dr \\ &= \frac{n!}{\alpha^{n+1}} \left[1 - e^{-\alpha b} \sum_{k=0}^n \frac{(\alpha b)^k}{k!}\right] \end{aligned}$$

Going back to the radial integrals for the potential

$$\begin{aligned} \int_0^r dr' r'^4 e^{-r'} &= 4! \left[1 - e^{-r} \left(1 + r + \frac{r^2}{2} + \frac{r^3}{6} + \frac{r^4}{24}\right)\right] \\ \int_0^r dr' r'^6 e^{-r'} &= 6! \left[1 - e^{-r} \left(1 + r + \frac{r^2}{2} + \frac{r^3}{6} + \frac{r^4}{24} + \frac{r^5}{120} + \frac{r^6}{720}\right)\right] \\ \int_r^\infty dr' r'^3 e^{-r'} &= 3! e^{-r} \left(1 + r + \frac{r^2}{2} + \frac{r^3}{6}\right) \\ \int_r^\infty dr' r' e^{-r'} &= 1! e^{-r} (1 + r) \end{aligned}$$

Plugging all that in, we find

$$\begin{aligned}
\Phi &= \frac{2}{3\epsilon_0} \frac{1}{64\pi} \left\{ \frac{24}{r} \left[1 - e^{-r} \left(1 + r + \frac{r^2}{2} + \frac{r^3}{6} + \frac{r^4}{24} \right) \right] P_0 \right. \\
&\quad - \frac{720}{5r^3} \left[1 - e^{-r} \left(1 + r + \frac{r^2}{2} + \frac{r^3}{6} + \frac{r^4}{24} + \frac{r^5}{120} + \frac{r^6}{720} \right) \right] P_2 \\
&\quad \left. + 6 e^{-r} \left(1 + r + \frac{r^2}{2} + \frac{r^3}{6} \right) P_0 - \frac{r^2}{5} e^{-r} (1 + r) P_2 \right\} \\
\Phi(r, \theta) &= \frac{1}{4\pi\epsilon_0} \left\{ \frac{1}{r} - \left(\frac{1}{r} + \frac{3}{4} + \frac{r}{4} + \frac{r^2}{24} \right) e^{-r} \right. \\
&\quad \left. - \left[\frac{6}{r^3} - \left(\frac{6}{r^3} + \frac{6}{r^2} + \frac{3}{r} + 1 + \frac{r}{4} + \frac{5r^2}{120} \right) e^{-r} \right] \frac{3 \cos^2 \theta - 1}{2} \right\}
\end{aligned}$$

for the exact solution with no restrictions on r . This reduces to the multipole expansion in part (a) for large r ,

$$\Phi(r \gg 1, \theta) \rightarrow \frac{1}{4\pi\epsilon_0} \left[\frac{1}{r} - 6 \frac{P_2(\cos \theta)}{r^3} \right]$$

Near the origin, we can use a small- r expansion of the radial part

$$\Phi(r, \theta) = \frac{1}{4\pi\epsilon_0} [A(r) - B(r)P_2(\cos \theta)]$$

and simplify the algebra using the fact that we have terms involving products of e^{-r} by the first few terms of e^r . For the coefficient of P_0 , this gives

$$\begin{aligned}
A(r) &= \frac{1}{r} \left[1 - e^{-r} \left(1 + r + \frac{r^2}{2} + \frac{r^3}{6} + \frac{r^4}{24} \right) \right] + \frac{1}{4} e^{-r} \left(1 + r + \frac{r^2}{2} + \frac{r^3}{6} \right) \\
&= \frac{1}{r} \left[1 - e^{-r} \left(e^r - \sum_{k=5}^{\infty} \frac{r^k}{k!} \right) \right] + \frac{1}{4} e^{-r} \left(e^r - \sum_{k=4}^{\infty} \frac{r^k}{k!} \right) \\
&= \frac{1}{r} \left[\frac{r^5}{5!} + \mathcal{O}(r^6) \right] e^{-r} + \frac{1}{4} \left[1 - \frac{r^4}{4!} e^{-r} + \mathcal{O}(r^5) \right] \\
&= \frac{1}{4} + \mathcal{O}(r^4)
\end{aligned}$$

and for the coefficient of P_2 , we have

$$\begin{aligned}
B(r) &= \frac{6}{r^3} \left[1 - e^{-r} \left(1 + r + \frac{r^2}{2} + \frac{r^3}{6} + \frac{r^4}{24} + \frac{r^5}{120} + \frac{r^6}{720} \right) \right] + \frac{r^2}{120} e^{-r} (1 + r) \\
&= \frac{6}{r^3} \left[1 - e^{-r} \left(e^r - \sum_{k=7}^{\infty} \frac{r^k}{k!} \right) \right] + \frac{r^2}{120} e^{-r} \left(e^r - \sum_{k=2}^{\infty} \frac{r^k}{k!} \right) \\
&= \frac{6}{r^3} \left[\frac{r^7}{7!} + \mathcal{O}(r^8) \right] + \frac{r^2}{120} \left[1 - \frac{r^2}{2!} + \mathcal{O}(r^3) \right] \\
&= \frac{r^2}{120} + \mathcal{O}(r^4)
\end{aligned}$$

Plugging these back into the expression for the potential

$$\Phi(r \ll 1, \theta) \approx \frac{1}{4\pi\epsilon_0} \left[\frac{1}{4} - \frac{r^2}{120} P_2(\cos \theta) \right]$$

(c) We need to sort out the dimensions that were hidden in the algebra above before we can calculate a numerical answer in engineering (SI) units

- the expression given for the charge density assumes a dimensionless variable r , a dimensionless density normalized to 1

$$\rho(r, \theta) = \frac{1}{64\pi} r^2 e^{-r} \sin^2 \theta, \quad \int \rho(r, \theta) r^2 dr d\Omega = 1$$

dimensionless multipole moments, and a potential with dimension $[1/\epsilon_0]$

- Charge density should have dimension $[Q/L^3]$ where Q, L stand for charge and length; we restore the correct dimension to the charge density by measuring r in units of a_0 and setting the total charge to $-e$ to describe an electron orbital in the hydrogen atom

$$\rho_e(r, \theta) = q_e |\psi|^2 = \frac{q_e}{a_0^3} \frac{1}{64\pi} \left(\frac{r}{a_0} \right)^2 e^{-r/a_0} \sin^2 \theta, \quad \int \rho_e(r, \theta) r^2 dr d\Omega = q_e \equiv -e$$

- Multipole moments q_{lm} and the potential Φ should have dimensions $[QL^l]$ and $[Q/(\epsilon_0 L)]$ respectively. In all the integrals above for the multipole moments, if we use a change of variable $r' \rightarrow u = r'/a_0$, the normalization factor $1/a_0^3$ for the charge density cancels the one from $d^3x' = a_0^3 d^3u$ but there's an extra factor from $r^l = a_0^l u^l$, so $q_{lm} \rightarrow a_0^l q_{lm}$. Terms in the multipole expansion of Φ have $1/r^{l+1}$ which is $a_0^{-(l+1)} (a_0/r)^{l+1}$. Thus, if we simply replace r with r/a_0 in the potential formula, we need an extra factor of a_0^{-1} in addition to $-e$ for the charge

$$\Phi(r \gg a_0, \theta) \rightarrow \frac{-e}{a_0} \frac{1}{4\pi\epsilon_0} \left[\frac{a_0}{r} - 6 \left(\frac{a_0}{r} \right)^3 P_2(\cos \theta) \right]$$

- For the potential near the origin where the nucleus is located, this looks like

$$\Phi_e(r \ll a_0, \theta) \approx -\frac{e}{4\pi\epsilon_0 a_0} \left[\frac{1}{4} - \frac{1}{120} \left(\frac{r}{a_0} \right)^2 P_2(\cos \theta) \right]$$

The interaction energy W for the charge distribution ρ_n of the nucleus in the presence of this electronic potential Φ_e is

$$\begin{aligned} W &= \int \rho_n(\vec{x}) \Phi_e(\vec{x}) d^3x \\ &= -\frac{e}{4\pi\epsilon_0 a_0} \left[\frac{1}{4} \int d^3x \rho_n(\vec{x}) - \frac{1}{120} \frac{1}{a_0^2} \int d^3x \rho_n(\vec{x}) r^2 P_2(\cos \theta) \right] \\ &= -\frac{e}{4\pi\epsilon_0 a_0} \left[\frac{1}{4} q_n - \frac{1}{120} \frac{1}{a_0^2} \int d^3x \rho_n r^2 \left(\frac{3 \cos^2 \theta - 1}{2} \right) \right] \end{aligned}$$

where $q_n = e$ is the nuclear charge for the hydrogen atom and the 2nd term is proportional to the quadrupole moment

$$Q_{zz} = \int d^3x \rho_n (3z^2 - r^2) = \int d^3x \rho_n r^2 (3 \cos^2 \theta - 1)$$

so we end up with

$$W = -\frac{e^2}{16\pi\epsilon_0 a_0} + \frac{e Q_{zz}}{960\pi\epsilon_0 a_0^3} = W_m + W_q$$

Plugging in the numerical values, $e = 1.6 \times 10^{-19}$ C, $\epsilon_0 = 8.854 \times 10^{-12}$ F/m, $a_0 = 5.29 \times 10^{-11}$ m, and assuming that the given value for the quadrupole moment is missing a factor of e with $Q_{zz}/e = 10^{-28}$ m², we find the monopole and quadrupole interaction terms as

$$\begin{aligned} W_m &\simeq -1.1 \times 10^{-18} \text{ J} = -6.8 \text{ eV} \\ W_q &\simeq 6.5 \times 10^{-28} \text{ J} \end{aligned}$$

and the corresponding frequency for the quadrupole term is

$$f = \frac{W_q}{h} \simeq 0.98 \text{ MHz}$$